

Basic Concepts and Linear Algebra

Optimization Problem Standard Form

min / max

$f(x)$

s.t.

$g_i(x) \leq b_i$ (inequality constraints)

$h_j(x) = c_j$ (equality constraints)

$x \in X$ (variable bounds)

Linear Algebra Fundamentals

- **Rank:** Maximum number of linearly independent rows/columns
- **Matrix multiplication:** $C = AB$, $c_{ij} = \sum_k a_{ik}b_{kj}$
- **Matrix inverse:** $AA^{-1} = I$
- **System of equations** $Ax = b$:
 - $m = n$, $\text{rank}(A) = n \rightarrow$ unique solution
 - $m < n \rightarrow$ infinite solutions
 - $\text{rank}(A) < \text{rank}([A|b]) \rightarrow$ no solution

Gradient and Hessian

- **Gradient:** $\nabla f(x) = \begin{bmatrix} \frac{\partial f}{\partial x_1} \\ \vdots \\ \frac{\partial f}{\partial x_n} \end{bmatrix}$
- **Hessian:** $\nabla^2 f(x) = \begin{bmatrix} \frac{\partial^2 f}{\partial x_i \partial x_j} \end{bmatrix}$
- **1st order Taylor:** $f(x) \approx f(x_0) + \nabla f(x_0)^T(x - x_0)$
- **2nd order Taylor:** $f(x) \approx f(x_0) + \nabla f(x_0)^T(x - x_0) + \frac{1}{2}(x - x_0)^T \nabla^2 f(x_0)(x - x_0)$

Optimality Conditions

- **First-order necessary:** $\nabla f(x^*) = 0$
- **Second-order sufficient:**
 - min problem: $\nabla^2 f(x^*)$ positive definite (PD)
 - max problem: $\nabla^2 f(x^*)$ negative definite (ND)

Convexity Decision Framework

Convex Set Decision

Definition: Set S is convex if:

- (1)
- (2)
- (3)
$$\forall x, y \in S, \forall \lambda \in [0, 1] : \lambda x + (1 - \lambda)y \in S$$
- (4)

Decision Process:

1. Check if for any two points in S , their line segment is in S
2. If YES $\rightarrow$ Convex Set
3. If NO $\rightarrow$ Non-convex Set

Examples of Convex Sets:

- Hyperplane: $\{x|a^T x = b\}$
- Halfspace: $\{x|a^T x \leq b\}$
- Ball: $\{x||x - c|| \leq r\}$
- Polyhedron: $Ax \leq b$

Convex Function Decision

Definition: Function $f(x)$ is convex if:

$$\forall x, y, \forall \lambda \in [0, 1] : f(\lambda x + (1 - \lambda)y) \leq \lambda f(x) + (1 - \lambda)f(y)$$

Test Methods:

1. **Definition test:** Check convex combination property
2. **1st order test:** $\forall x, y : f(y) \geq f(x) + \nabla f(x)^T(y - x)$
3. **2nd order test:** $\nabla^2 f(x) \succeq 0$ (PSD)

PSD Test Methods:

- All eigenvalues ≥ 0
- Leading principal minors ≥ 0
- $x^T A x \geq 0 \forall x$

Convex Programming and Linear Programming

Convex Programming Decision

Problem is Convex if:

For Minimization:

- 1. Objective function $f(x)$ is convex
- 2. Feasible region is convex:
 - Inequality constraints $g_i(x) \leq b_i$ where g_i convex
 - Equality constraints $h_j(x) = c_j$ where h_j linear

For Maximization:

- 1. Objective function $f(x)$ is concave
- 2. Feasible region is convex

Linear Programming

Standard Form:

$$\begin{aligned} \min \quad & c^T x \\ \text{s.t.} \quad & Ax = b \\ & x \geq 0 \end{aligned}$$

(5)
(6)
(7)

Key Terms:

- **Basic solution:** Solution with n basic variables
- **Basic feasible solution (BFS):** Basic + feasible
- **Degeneracy:** Basic variable = 0
- **Extreme point:** Vertex of polyhedron
- **Extreme ray:** Unbounded direction

Simplex Method Steps

- 1. Find **initial BFS**
- 2. **Optimality test:** All reduced costs ≤ 0 ?
- 3. **Choose entering variable:** Most negative reduced cost
- 4. **Minimum ratio test:** Find step size
- 5. **Choose leaving variable:** Update basis
- 6. Repeat until optimal

Special Problems and Practical Formulas

Transportation Problem

$$\min \sum_i \sum_j c_{ij} x_{ij}$$

(8)

$$\text{s.t.} \quad \sum_j x_{ij} \leq s_i \quad (\text{supply})$$

(9)

$$\sum_i x_{ij} \geq d_j \quad (\text{demand})$$

(10)

$$x_{ij} \geq 0$$

(11)

Infeasibility condition: $\sum s_i < \max\{d_j\}$
(Total supply less than maximum demand)

Absolute Value Linearization

$$\min |x - a| \quad \rightarrow \quad \min z$$

(12)

$$\text{s.t.} \quad z \geq x - a$$

(13)

$$z \geq -(x - a)$$

(14)

Set Properties

- Set $X = \{x \in [a, b]^n : \sum x_i \geq c\}$
- **Bounded:** $[a, b]$ are finite intervals
 - **Closed:** Defined by closed inequalities
 - **Convex:** Intersection of box constraints + linear constraint

Common Convex Functions

- x^k ($k \geq 1, x \geq 0$): convex
- $\ln(x)$ ($x > 0$): concave
- e^x : convex
- $x^2 + y^2$: convex
- $\max\{f_1(x), f_2(x)\}$: convex if f_1, f_2 convex
- $|x|$: convex

Decision Timing Concepts

- **Here-and-now:** Decision made before uncertainty is realized
- **Wait-and-see:** Decision made after uncertainty is realized

Key Calculations from Past Exams

- **Gradient example:** $f(x_1, x_2) = \sqrt{x_1 - 1} + (x_2 - 1)^{1/4}$ at $(3, 2)$

$$\nabla f = \begin{bmatrix} \frac{1}{2\sqrt{2}} \\ \frac{1}{4} \end{bmatrix}$$

- **Hessian example:** $f(x_1, x_2) = 2(x_1 - 3)^2 + 6x_1x_2 + 2(x_2 + 3)^2 + 6x_1 + 2x_2$ at $(2, 3)$

$$H = \begin{bmatrix} 4 & 6 \\ 6 & 4 \end{bmatrix}$$

Important Abbreviations and Symbols

- **LP:** Linear Programming
- **NLP:** Nonlinear Programming
- **BFS:** Basic Feasible Solution
- **PSD:** Positive Semidefinite ($\succeq 0$)
- **PD:** Positive Definite ($\succ 0$)
- **s.t.:** subject to
- $\forall$: for all
- $\exists$: there exists